IDX

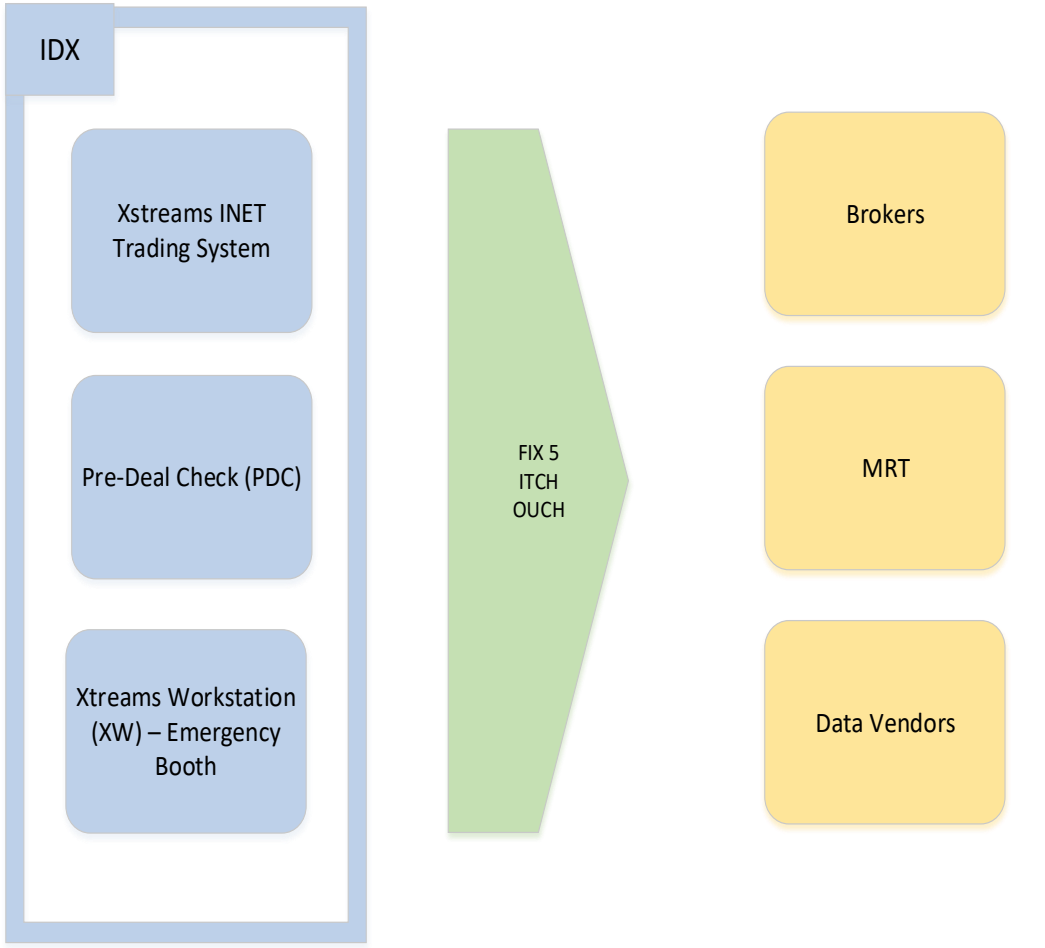
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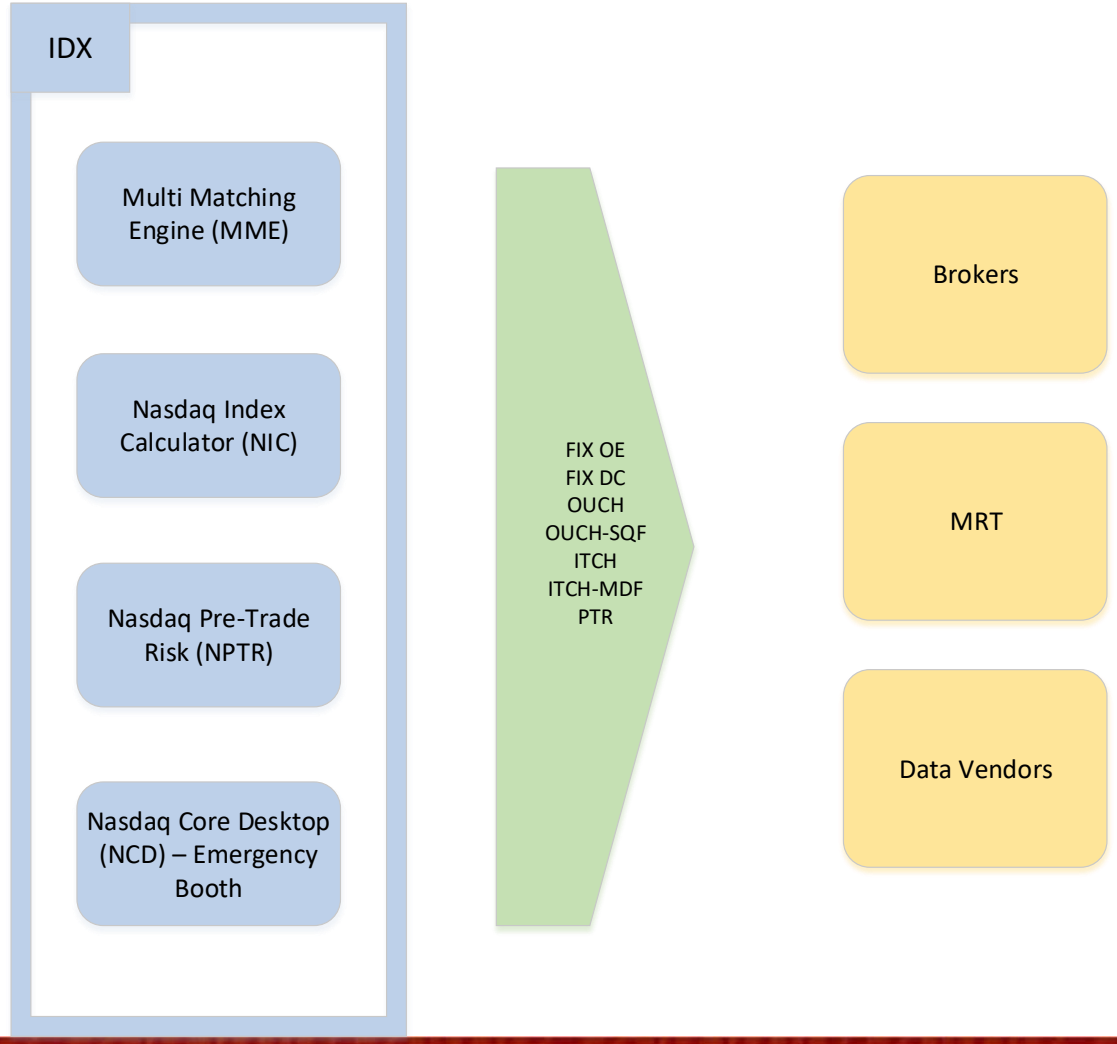
Development Scope for Brokers and Data Vendors in PSPP

IDX Trading System Changes

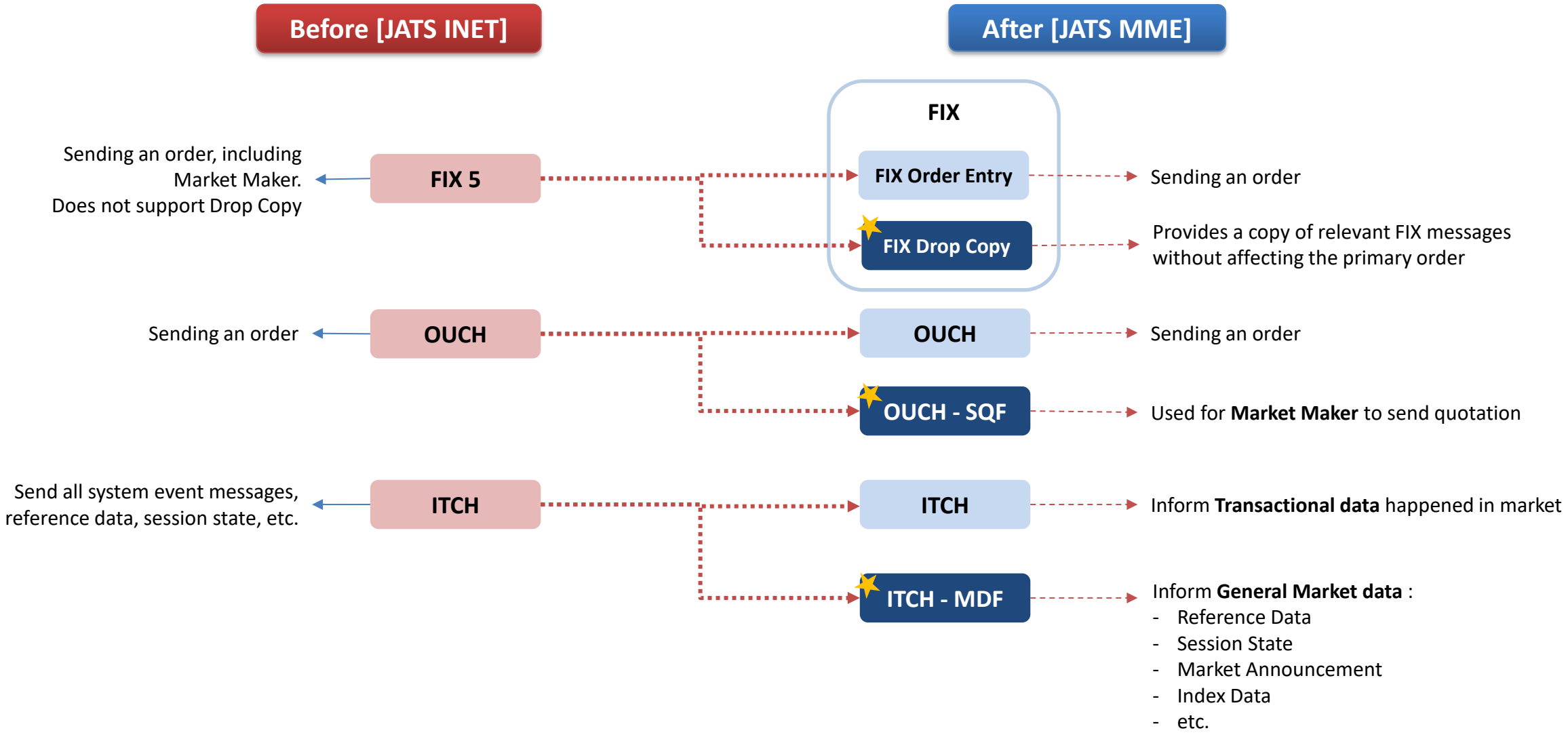
- Current Systems



- New Systems



Protocols in JATS (1)



New Protocols in JATS

OUCH – SQF

This protocol is **critical for Market Makers** or or high-frequency traders

- **Designed for market makers**, enabling to maintain tight spreads and provide liquidity information in real time
- **Low Latency** for quotes update in high-frequency trading
- Able to send and update **large volumes of quotes** (bulk quoting) efficiently
- Able to send **multiple quotations** up to 10 levels

ITCH – MDF

Distributes complementary market data **that not contained in ITCH**.

List of information broadcasted :

- Time Message – Timestamps
- Reference Data Message
- Event and State Change Message
- Price Message
- Market Announcement

FIX Drop Copy

- Provides copies of trade-related messages for back-office or **monitoring use**
- Doesn't handle order execution directly, but **copies of messages** like execution reports and order updates
- **One-way** copy of data

Connecting to JATS

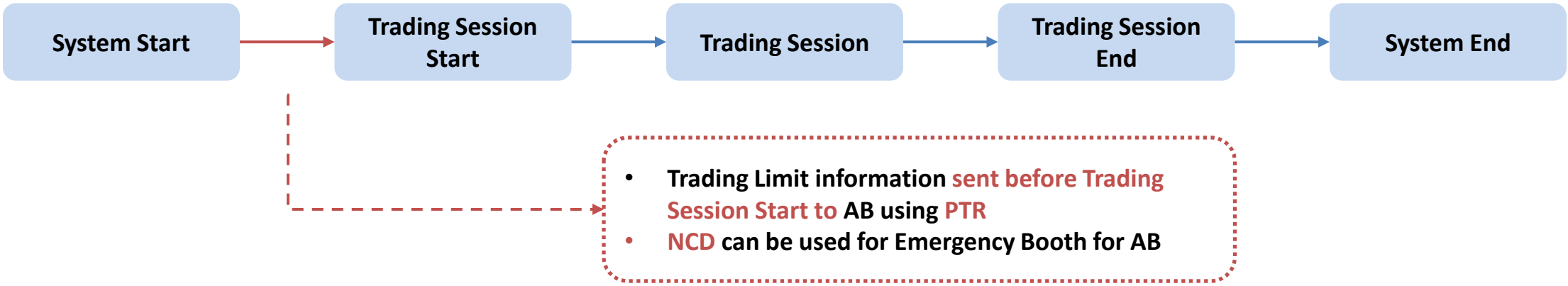
Connecting to FIX OE and FIX DC

- Define IP Address and Port
- Set the CompID (SenderCompID and TargetCompID)
- Send a Logon Message (**MsgType=A**) with UserCode and Password

Connecting to OUCH, OUCH - SQF, ITCH and ITCH - MDF

- All of these protocols connected to JATS via **SoupBinTCP**
- Establish a **TCP socket connection** and define IP Address and Port
- **SoupBinTCP** sends an **unsolicited login accept** message with SoupUserCode, Password and seqnum = 1

All Protocol in JATS (2)



PTR

- Controls that are applied **before a trade is executed**
- Ensure that traders and systems comply with certain risk limits before trades reach the market
- Used to **track their Trading Limit** in real-time during trading sessions

NCD

- Act as Emergency Booth for brokers
- Can be used for **emergency withdraw orders**

Message Flow Examples

Before			After				
FIX 5	OUCH	ITCH	FIX 5	OUCH	ITCH	OUCH-SQF	ITCH-MDF
-	-	System Event (S)	-	-	-	-	System Event (S)
Logon (A)	Login Request (SoupBinTCP)	Login Request (SoupBinTCP)	Logon (A)	Login Request (SoupBinTCP)	Login Request (SoupBinTCP)	Login Request (SoupBinTCP)	Login Request (SoupBinTCP)
New Order Single (D)	New Order (O)	-	New Order Single (D)	New Order (O)	-	-	-
Execution Report (8)	Accepted Message (A)	Add Order Message (A)	Execution Report (8)	Accepted Message (A)	Add Order Message (A)	-	-
Execution Report (8)	Executed Order Message (E)	Order Executed Message (E)	Execution Report (8)	Executed Order Message (E)	Order Executed Message (E)	-	-
Order Cancel/Replace Request (G)	Replace Order Message (U)	Order Replace Message (U)	Order Cancel Replace Request (G)	Replace Order (U)	Modify Order Message	-	-
Mass quotes (i)	-	-	Mass quotes (i) *for Multi Quotes - TBD	-	-	SQF Quote Block (Q)	-
Mass Action Request (CA)	-	-	-	-	-	SQF Cancel Request (V)	-
Trade Capture Report (AE)	-	-	Trade Capture Report (AE)	-	-	-	-
-	-	Circuit Breaker Message (C)	-	-	-	-	Circuit Breaker Message (C)

Summary of JATS Protocol

Protocols	Use Case	Connection Method	Message Flow	Notes
FIX Order Entry (FIX 5.0 using OrderEntry gateway)	Placing, modifying, and canceling orders in real-time trading systems	TCP/IP	Client sends New Orders, Modifications, Cancellations, receives Execution Reports	Available in current JATS as FIX5.0
FIX Drop Copy (FIX 5.0 using DropCopy gateway)	Captures a duplicate of trading activity for monitoring order using other system	TCP/IP	Client receives copies of execution reports, trade confirmations, etc.	Not Available in current JATS
OUCH	Low-latency order Placing, modifying, and canceling orders in real-time trading systems	SoupBinTCP over TCP/IP	Client sends order placement and modification messages, receives execution reports	Available in current JATS as OUCH
OUCH – SQF (Specialized Quote Interface)	Low-latency market-making, updating bid/ask quotes for multiple quotations	SoupBinTCP over TCP/IP	Client sends and receives quotes for market-making	Not Available in current JATS
ITCH	Real-time market data feed, full depth of the order book	SoupBinTCP over TCP/IP	Server sends real-time transactional data updates on trades and order book	Available in current JATS as ITCH
ITCH - MDF	Real-time consolidated market data from Nasdaq markets	SoupBinTCP over TCP/IP	Server sends Market Data Feed (MDF) with consolidated market data	Not Available in current JATS (Currently, messages are combined in ITCH)
PTR	PTR ensures trades comply with risk limits and Track trading limit available	Web Based	Risk checks on order submissions (e.g., size, price, exposure limits) for order execution or rejection	
NCD	Act as emergency booth for Traders (**withdraw Only**)	Web Based	Continuous updates on available TL for trading purposes	

IDX

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Thank You